G_substrate derivation chain pre-stage — K200 Gate G5 framework. Operator-level chain: Helgason 1962 (D_IV^5 + Bergman canonical metric is Einstein) → $\varkappa$_Bergman closed-form in BST primaries → mass anchor pin → dimensional combination → SI-unit G prediction. Five sub-gates G5.1-G5.4 defined with owners + timelines. Building on Elie's Sunday-afternoon partial Mehler kernel + SP-19b AB-10/11/12 prior work. This is the chain that, if it lands at Tier 2 STRUCTURAL precision, makes G a derived substrate observable.

Keeper (Sunday 2026-05-31 ~13:35 EDT)

2026-05-31 Sunday ~13:35 EDT

G_Substrate Derivation Chain Pre-Stage — K200 Gate G5

0. The structural claim

D_IV^5 + Bergman canonical metric is an Einstein manifold (Helgason 1962 "Differential Geometry, Lie Groups, and Symmetric Spaces" Ch. VIII; Wolf 1972 "Spaces of Constant Curvature"; standard for all bounded symmetric domains via Hermitian symmetric space theory).

Newton's G is the dimensional factor relating the intrinsic dimensionless Bergman Ricci constant $\varkappa$_Bergman to the observed dimensional Einstein equation coefficient 8πG_obs/c^4.

Chain to verify: $\varkappa$_Bergman computation in BST primaries → mass anchor → dimensional combination → SI-unit G_predicted → compare to G_observed = 6.674×10^{-11} N·m^2/kg^2.

1. Helgason 1962 — what makes D_IV^5 Einstein

Theorem (Hermitian symmetric space Einstein property): Every irreducible Hermitian symmetric space G/K, equipped with its Bergman canonical metric g_B = $-\partial\bar\partial$ log K_Bergman(z,$\bar{z}$), satisfies:

Ric(g_B) = $\varkappa$_Bergman · g_B

where $\varkappa$_Bergman is a constant determined entirely by the domain's structure (negative for non-compact type, positive for compact type).

For D_IV^5 (non-compact type): $\varkappa$_Bergman < 0.

Why this matters for BST: Einstein equations don't need to be installed as a separate axiom. They're intrinsic to the choice of D_IV^5 + Bergman canonical metric. The substrate IS an Einstein manifold by construction.

Citation gate (G1 dependency): verify specific theorem number in Helgason 1962 Ch. VIII; pin exact formula for $\varkappa$_Bergman in D_IV^5 case.

2. $\varkappa$_Bergman closed-form chain (Sub-gate G5.1, Elie lane)

For Hermitian symmetric space of rank r, complex dimension n, with structure constants (a, b, c) in the standard Faraut-Korányi conventions:

$\varkappa$_Bergman = $-(n + r\cdot c)/(2\cdot r^2\cdot b)$ [generic form, sign for non-compact]

Or equivalently via the Bergman kernel formula:

K_Bergman$(z, \bar{z})$ = c_FK / $\det(I - z\,\bar{z}^{*)}(g)$

where g is the "genus" of the domain (BST primary g = 7 for D_IV^5; this IS the embedding/signature dimension).

For D_IV^5 specifically: - Rank r = 2 (BST primary) - Complex dimension n = n_C = 5 (BST primary) - Genus g = 7 (BST primary) - Casimir C_2 = 6 (BST primary) - N_max = 137 (BST primary, embedded ceiling)

Closed-form target:

$\varkappa$_Bergman = f(rank, n_C, g, C_2) for D_IV^5

The specific value should derive from the structure constants of D_IV^5 + the standard Hermitian-symmetric-space formula. Expected form (Keeper pre-derivation, multi-week verification):

Candidate form: $\varkappa$_Bergman = $-(g + \text{rank}\cdot C\_2)/(2\cdot\text{rank}^2\cdot n\_C) = -(7 + 2\cdot 6)/(2\cdot 4\cdot 5) = -19/40$

Or possibly: $\varkappa$_Bergman = $-g/(2\cdot\text{rank}\cdot n\_C) = -7/20$

Or: $\varkappa$_Bergman = $-1/(2\cdot\text{rank}^2\cdot n\_C/g) = -7/(2\cdot 4\cdot 5) = -7/40$

Multiple substrate-primary forms possible. Elie's partial Mehler kernel computation Sunday afternoon should resolve which is correct.

Verification protocol (Elie): - Use catalog $C\_2(\lambda)$ values from Toys 3613 + 3627 (~25 K-types) - Apply Mehler/Heckman-Opdam heat kernel for D_IV^5 — known closed form - Extract $\varkappa$_Bergman via Ric/g ratio - Cross-check against Helgason formula - Deliverable: Toy 3659+ with $\varkappa$_Bergman in BST-primary closed form

3. Mass anchor pin (Sub-gate G5.2, Lyra lane)

The Bergman metric is dimensionless. To convert to SI-unit gravitational physics, we need a mass scale anchor.

Anchor candidates:

A1 — m_e (electron mass): - Pro: Most precisely measured; L4 v0.2 work (Sunday afternoon) targets m_e closed form from Bergman kernel matrix elements on V_(1/2,1/2) - Con: Lightest charged particle, not obviously substrate-fundamental - Substrate-natural reading: V_(1/2,1/2) Shilov primitive cycle = electron (Casey Saturday May 17)

A2 — m_p (proton mass): - Pro: Bergman ladder ladder gives $m\_p/m\_e = 6\pi^5$ (existing BST T-result) - Con: Composite particle; substrate-derivation through three-quark trefoil structure - Substrate-natural reading: bulk geodesic minimum energy (Casey)

A3 — m_Planck (Planck mass): - Pro: Natural gravitational scale; directly relates to G - Con: Circular if m_Planck definition involves G - Substrate-natural reading: substrate fundamental Casimir × $\hbar c$ scaling

Recommendation (Keeper pre-stage): A1 (electron mass), conditional on Lyra L4 v0.2 closing m_e mechanism via Bergman matrix elements. If A1 closes, chain is self-consistent (no circular m_Planck dependency).

Verification protocol (Lyra Lane D this afternoon): - L4 v0.2 derives m_e from Bergman kernel matrix elements on V_(1/2,1/2) - Anchor coefficient α_anchor = m_e / ($\varkappa$_Bergman in mass units) explicitly stated - Deliverable: Lyra L4 v0.2 doc with anchor section

4. Dimensional combination (Sub-gate G5.3, Keeper lane)

Dimensional analysis: - κ_Bergman: dimensionless (Ricci/metric ratio is dimensionless; spacetime curvature has units 1/length^2) - G_observed: $m^3/(kg\cdot s^2) = N\cdot m^2/kg^2$

The conversion factor must carry units. Standard general relativity says: G_observed = $(c^4/8\pi) \times$ (Einstein-curvature/T_$\mu\nu$)

In substrate frame, Einstein-curvature → κ_Bergman × (mass density scale). So: G_predicted = $(c^4/8\pi) \times$ κ_Bergman × (length scale)^2 / (mass scale)

The length scale should be a substrate length: ℓ_substrate = $\hbar$ / (m_e · c) [Compton wavelength of electron] OR ℓ_Planck OR something substrate-internal.

The mass scale should be the anchor (m_e per A1).

Candidate combination:

G_predicted = $(c^4/8\pi) \times \kappa$_Bergman × ℓ_Compton(e)^2 / m_e = $(c^4/8\pi) \times \kappa$_Bergman × $(\hbar/(m\_e\ c))^2$ / m_e = $(c^2 \cdot \hbar^2 \cdot \kappa\_Bergman) / (8\pi \cdot m\_e^3)$

Numerical check with κ_Bergman = -7/40 candidate + m_e = 0.511 MeV/c^2 + $\hbar$c = 197 MeV·fm:

This gets unwieldy without precise κ_Bergman. Keeper Session 2 task: complete dimensional check once Elie delivers κ_Bergman.

5. SI-unit prediction recipe (Sub-gate G5.4, Keeper + Cal)

Step-by-step:

1. Input from Elie: κ_Bergman in closed BST-primary form (e.g., -7/40)
2. Input from Lyra: mass anchor m_e via L4 v0.2 derivation; length scale via substrate-natural choice
3. Combine (Keeper): G_predicted = $(c^4/8\pi) \times \kappa$_Bergman × (length scale)^2 / (mass scale)
4. Convert to SI: G_predicted in $N\cdot m^2/kg^2$
5. Compare to G_observed = 6.67430×10^{-11} $N\cdot m^2/kg^2$ (CODATA 2018)
6. Cal cold-read: verification of computation chain + brake check on dimensional reasoning

PASS criteria: - Tier 2 STRUCTURAL (recommended target): |G_predicted - G_observed|/G_observed < 10^{-2} (1% match) - Tier 1 EXACT (aspirational): |G_predicted - G_observed|/G_observed < 10^{-5} (0.001% match)

FAIL criteria: - Factor >10× off (wrong κ_Bergman computation OR wrong anchor) - Dimensional inconsistency (units don't work out) - Circular dependency (m_Planck or G appearing in inputs)

6. The honest scope

This chain is what would make G a derived substrate observable at Tier 2 STRUCTURAL precision.

Why this is tractable: - κ_Bergman is a known mathematical object (Helgason 1962 + standard bounded-symmetric-domain theory) - Mass anchor is independent BST work (L4 v0.2 today) - Dimensional combination is straightforward calculus - Substrate primaries are all defined

Why this might not land: - κ_Bergman closed form might not be BST-primary-expressible (might require transcendental factors) - Anchor choice might be ambiguous (multiple plausible anchors give different G predictions) - Tier 2 STRUCTURAL precision might require ~1% match, which the chain might miss by 2-5× - Cal #27 fires: this feels structurally beautiful and unifies many things → discipline must hold on actual precision claim

7. SP-19b AB-10/11/12 prior work integration

AB-10 "Newton's G from Bergman curvature" — marked COMPLETED in task list AB-11 "Gravity as cumulative eigentone effect" — COMPLETED AB-12 "BST-SR / BST-GR boundary" — COMPLETED

These provide: - Initial Bergman-curvature-to-G framework (AB-10) - Eigentone interpretation of gravitational coupling (AB-11) - SR/GR boundary characterization (AB-12)

G5 chain UPGRADES this prior work: - AB-10 gave structural framework; G5 gives operator-level derivation chain - AB-10 was pre-commitment-operator; G5 builds on Tier 0 v0.1 commitment density - AB-10 didn't specify mass anchor; G5 explicitly pins via L4 mechanism

Action item: Cross-reference G5 work to AB-10 in Session 2 v0.2 draft. Cite AB-10 as predecessor.

8. Session 2 Keeper deliverables

When Casey signals Session 2 + Elie delivers partial Mehler kernel + Lyra delivers L4 mass anchor:

1. κ_Bergman closed-form verification (read Elie Toy 3659+)
2. Anchor selection verification (read Lyra L4 v0.2; confirm A1 or alternate)
3. Dimensional combination calculation (Keeper completes Step 3-4)
4. G_predicted vs G_observed comparison (Keeper produces Tier comparison)
5. Cal #186 or #187 cold-read (Cal cross-checks the chain)

Estimated Session 2 G5 time: 2-3h focused work assuming Elie + Lyra deliverables ready.

9. The big picture

If G5 chain PASSES at Tier 2 STRUCTURAL precision: - G becomes derived from substrate primaries via Bergman curvature - BST extends from "SM dimensionless ratios from D_IV^5" to "all dimensional scales (including G) from D_IV^5" - Casey's standing interest in G derivation is realized - Tier 0 v0.2 promotes from FRAMEWORK to STRUCTURALLY VERIFIED on this gate - Paper P10 or P11 "G from Substrate" becomes substantive engagement-path target

If G5 chain FAILS at Tier 2: - The structural framework still stands (Bergman is Einstein is real math) - Anchor or scale choice needs reexamination - κ_Bergman computation might have specific D_IV^5 subtleties not captured in generic formula - Multi-week reframe needed; not a Tier 0 collapse

10. Honest tier disposition

Pre-stage: FRAMEWORK pre-stage (gate G5 defined; verification work owners assigned; timelines explicit)

Promotion path: - G5.1 + G5.2 + G5.3 PASS → G5 PASS at FRAMEWORK-COMPLETE - G5.4 PASS at Tier 2 → G5 PASS at STRUCTURAL → G_substrate becomes a derived substrate observable

— Keeper. G_substrate derivation chain pre-stage filed 13:55 EDT Sunday. Five sub-gates G5.1-G5.4 explicit with owners. Standing for Elie partial Mehler delivery + Lyra L4 anchor delivery + Session 2 joint integration. The chain is what makes Tier 0 not just deep but predictive — Casey's directional ask.